FEATURE-GUIDED BLIND FACE RESTORATION WITH GAN PRIOR

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ABSTRACT

Blind face restoration (BFR) aims to restore high-quality face images from inputs with complex degradation, which is key to extensive applications. Existing methods usually learn a black-box mapping to achieve the goal, which however often produce over-smoothed results. This work proposes a novel feature-guided framework via leveraging prior from a pre-trained Generative Adversarial Network (GAN) model to recover reasonable textures. Furthermore, we design a feature fusion module to guide the generator with low-level spatial content information in the degraded input, for the sake of holding the consistency on both facial structure and background details. The proposed method can be seamlessly integrated with a learned GAN through a simple yet effective principle to recover realistic results under complex degradation circumstances. Extensive comparisons demonstrate the superiority of our strategy over other state-of-the-art methods in terms of restoration quality and training cost.

***Index Terms*—** Blind face restoration, feature guidance, feature fusion, GAN prior

1. INTRODUCTION

Portraits play a crucial role in our daily life, yet often suffer from degeneration owing to complex degradation such as low resolution, blur, noise, and compression. Face image restoration has been rapidly developing due to its widespread applications, which aims to restore high-quality (HQ) face images from corresponding low-quality (LQ) observations. Despite huge research attention being received over recent years, various hurdles, including complex degraded patterns and sensitivity of the human visual system to minor visual mistakes in synthesized images, require further consideration in practice. Recently, most existing CNN-based approaches have accomplished excellent progress on face restoration tasks, assuming that the degradation is certain and fixed, e.g., super-resolution, denoising, deblurring, and compression artifact removal. Nevertheless, these task-specific methods are usually sensitive to degeneration patterns with poor generalization ability in handling real-world images with diverse and mixed degradation.

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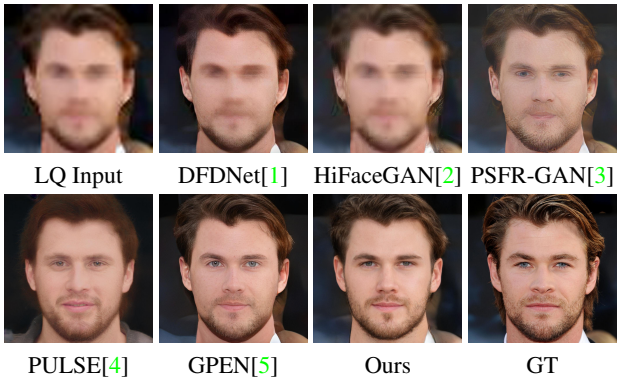


Fig. 1: A visual comparison with state-of-the-art methods.

Blind face restoration (BFR) is in nature ill-posed. In other words, many HQ images can correspond to the same LQ image, and numerous BFR mechanisms attempted to solve the problem from different views. For instance, exemplar-based [6, 7, 8] and facial component dictionary-based [1] methods incorporate reference images as additional guidance to restore clear facial structure, preserving the lineament but hardly restoring the photo-realistic textures. PSFR-GAN [3] formulate the restoration of LQ face images as a multi-scale progressive restoration procedure through semantic-aware style transformation. GAN prior based methods [4, 5, 9] adopt pre-trained GAN prior to cope with the complex degradation, such as StyleGAN [10, 11]. However, the low-dimensional latent codes are insufficient to restore all details, especially for highly-degraded images due to information loss. Thus, these approaches often suffer from the inconsistency between foreground and background contents. Another point that needs to be mentioned is the expensive cost for training GAN models.

In this work, we focus on leveraging the diverse priors of pre-trained GANs for the task in various degraded scenes. As discussed in [12], the encoder in the GAN inversion task favours to hold the primary information of a domain, and the lost part for the face domain mainly comes from background details. Based on this analysis and experimental observations, we incorporate feature fusion module to further guide generator to reconstruct the ignored image-specific information. Utilizing the feature fusion module can greatly help us

achieve comparable results in the BFR task with significantly less training cost than other GAN prior methods. In addition, we validate our approach through quantitative and qualitative experiments and exhibit that our strategy accomplishes state-of-the-art results for the task of BFR. Concretely, our contributions can be summarized as follows:

- We straight employ a pre-trained GAN to restore high-quality textures for degraded facial images to cope with sophisticated degradation, achieving competitive results with much less training cost than other competitors.
- We present a feature-guided framework for boosting the global structure and background textures using diverse generative facial priors with the guidance of structural information, by which our method can produce visually striking and realistic results.
- We conduct extensive experiments to evaluate our framework. Both qualitative and quantitative results demonstrate the superiority of our method over other state-of-the-art BFR methods. Our framework can be simply applied to other pre-trained GAN models.

2. RELATED WORK

Due to limited space, this section will briefly review representative methods in the fields of blind face restoration and GAN prior for face restoration, which are closely related to ours.

Blind face restoration aims to predict the restoration based on observed samples without knowing degradation information. Most existing works deal with the BFR task via adopting a reference-based strategy or carefully designed framework. As a representative, HiFaceGAN [2] introduced collaborative suppression and replenishment framework to recover face details progressively. Reference-based methods [6, 7, 8] employed the textures from reference images to improve restoring quality. DFDNet [1] proposed multi-scale component dictionaries to release the requirement of identity-belonging referenced images. Nonetheless, the variations in poses and expressions between the reference and LQ images will inevitably affect the performance of these methods, limiting their applicability. Besides, abundant research in unconditional generative image modeling has been carried out since the preliminary work of GAN was developed [13]. Thanks to its excellent ability to generate photo-realistic images, numerous works utilized pre-trained face GAN models with diversity prior to reconstruct faithfulness results for complex degradation in real-world scenarios. PULSE [4] directly optimized the latent code of StyleGAN [11] by constraining a down-scaling loss between input and output. mGANprior [14] sought multiple codes to improve the reconstruction quality. However, limited/reduced dimensions of latent space make it challenging to recover the target facial structure faithfully. GPEN [5] used an encoder-decoder framework with skip connection to somewhat save the information loss during mapping to latent space and redesigned the GAN blocks of StyleGAN for coping with complex degradation, which is computationally intensive to retrain the generator that can synthesize realistic photos.

In contrast, our proposed scheme adopts a feature-guided module to provide additional spatial guidance for restoration, and utilizes GAN prior with fine facial textures and color information to remedy the degraded input. Besides, our framework achieves competitive results in comparison with the state-of-the-art methods at a considerably cheaper cost by keeping the pre-trained GAN unchanged for face restoration.

3. PROPOSED METHOD

3.1. Overall Design

Pre-trained GAN models on nature images contain rich and realistic texture and shape priors. However, directly seeking optimal latent codes for degraded images is challenging and time-consuming. For practical use and possibly broadening applicable scenarios, this work concentrates on how to reduce the cost of employing/training GAN prior for image restoration tasks. The overall architecture of our network is depicted in Fig. 2(a). As can be seen, the whole network consists of an encoder and a generator (pre-trained StyleGAN2 [11]). Before going into details, we briefly introduce the problem set with basic notations. Let I_L denote a given LQ image, while I_H designates its corresponding HQ version. The encoder learns to transfer I_L from the image space into the latent space via $\mathcal{E}(I_L) \rightarrow \hat{W}$. Then, we employ a learned generator of GAN to restore image $\hat{I}_H$ through $\mathcal{G}(\hat{W}) \rightarrow \hat{I}_H$. As suggested in [15] that customizes a lightweight and efficient network for the GAN inversion task, we adopt feature maps at 3 different levels by a pyramid network. A shared hierarchical mapping module is introduced to predict latent codes at each feature level. Moreover, we design a feature-guided strategy for easier utilizing priors, which can reduce the complexity of GAN inversion optimization by aggregating the feature maps from the encoder to guide the generator. Only one layer of the generator is performed the fusion operation, which works sufficiently well in our experiments. To be more convincing, the ablation study will offer several comparisons on specific fusion strategies to support the claim. The feature fusion module can instruct the generator to recover high-quality results with high fidelity to input and low time cost (ours vs. GPEN [5]: 20 hours vs. $\sim$ 14 days) to reach training convergence. In what follows, we will detail the proposed feature-guided fusion module together with involved loss terms. We will show in experiments that such a simple strategy can straightly take advantage of existing pre-trained GAN models without specific retraining, and effectively restore degraded images.

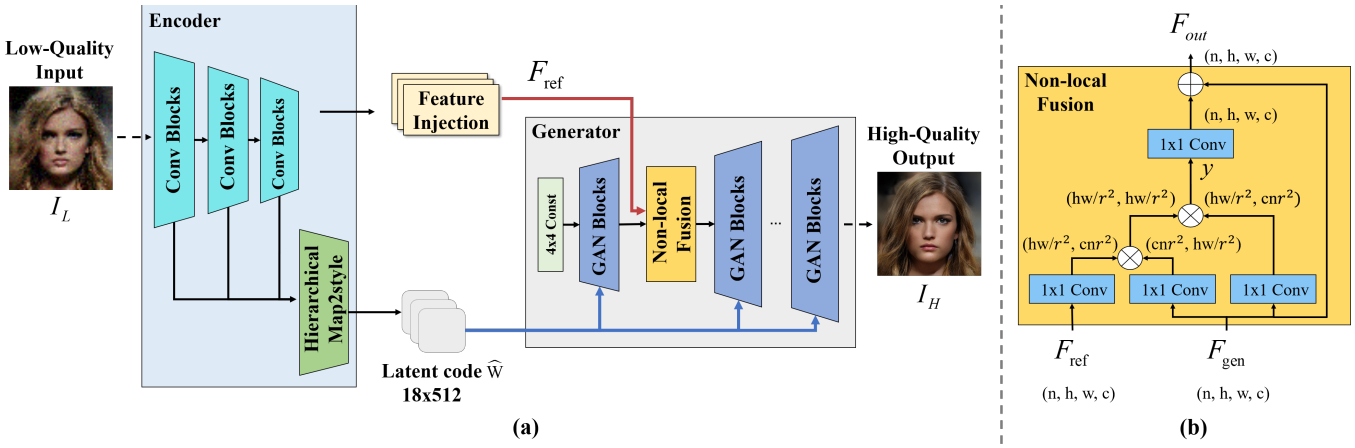


Fig. 2: (a) **Overview of our framework**. Given a degraded face image I_L as input, our framework first uses an encoder to extract its feature map F_{ref} and predict the corresponding latent code $\hat{W}$. After that, the $\hat{W}$ is fed into the generator. A Non-local fusion module is used to aggregate feature maps to synthesis the HQ image $\hat{I}_H$. (b) **Non-local fusion module**. The feature maps F_{ref} and F_{gen} are respectively from the encoder and generator with size $32 \times 32 \times 512$. $\bigotimes$ denotes the matrix multiplication, while $\bigoplus$ stands for the element-wise sum.

3.2. Feature-guided Fusion Module

As illustrated in Fig. 2(b), the non-local fusion module aggregates the convolutional features dynamically by considering the non-local relation between the LQ and HQ image to handle the mismatched pixel-level feature maps. The non-local operation is defined as follows:

$$y = \frac{1}{\tau} f\left(F_{ref}, F_{gen}\right) \otimes g\left(F_{gen}\right), \tag{1}$$

where τ is the normalized factor set by the number of positions in feature maps. Function f computes the relationship between F_{ref} and F_{gen}, producing an attention map by the softmax. Prior to feeding the features into f, 1×1 convolutions are adopted to reshape them to $(hw/r^2, cnr^2)$, where n, c, h and w represent the batch size, number of channels, height, and width of the input feature maps, respectively. Moreover, r stands for the reduction factor. As for function g, it computes a feature representation from the generator. Then, we can obtain fused feature maps via:

$$F_{out} = \mathcal{W}y \oplus F_{gen}, \tag{2}$$

where $\mathcal{W}$ stands for the 1×1 convolution operation. Then we reshape $\mathcal{W}y$ to (n, h, w, c) and combine it with F_{gen}.

This paper uses dot production as the pairwise function f to calculate the similarity for the sake of avoiding additional parameters, which yields:

$$f\left(F_{ref}, F_{gen}\right) = \exp({F_{ref}}^T F_{gen}). \tag{3}$$

Different from the original non-local structure, we alternatively use F_{ref} and F_{gen} as the inputs. Furthermore, we replace the Batch Normalization (BN) with Instance Normalization (IN) to service our task and initialize the scale parameter of the IN layer to zero, ensuring that the initial state of the non-local block is an identity mapping while maintaining initial generator behavior. As a result, we utilize the non-local module to integrate the features with a progressive learning strategy, which considerably reduces the training complexity of the encoder.

3.3. Loss Function

To train our encoder and fusion module, the following loss terms are used. We adopt the mean square error (MSE) to enforces the recovery $\hat{I}_H$ to be close to the ground truth I_H in the image space as follows:

$$\mathcal{L}_{\text{mse}} = \|\hat{I}_H - I_H\|_2^2, \tag{4}$$

where $\|\cdot\|_2$ stands for the ℓ_2 norm. In addition, we utilize the LPIPS loss [16] to measure the perceptual distance as follows:

$$\mathcal{L}_{\text{perc}} = \mathcal{L}_{\text{LPIPS}}(\hat{I}_H, I_H), \tag{5}$$

which is to further ensure the recovery looks like the ground truth in the feature space, and thus improving the visual/perceptual quality of the generated image. To constrain the encoded latent $\hat{W}$ in the manifold of pre-trained StyleGAN, we sample latent code W as the ground truth. Simply adopting the ℓ_2 norm as the latent recovery loss to encourage the encoder output latent vectors close to ground truth in the latent space as follows:

$$\mathcal{L}_{\text{w}} = \|\mathcal{E}(I_L) - W\|_2^2. \tag{6}$$

Considering the face restoration task is sensitive to the identity of the LQ input, we utilize the face recognition model to act as a regularization term, which can be expressed as follows:

$$\mathcal{L}_{\text{id}} = 1 - < \mathcal{F}(\hat{I}_H), \mathcal{F}(I_H) >, \tag{7}$$

where $\mathcal{F}$ is the pre-trained ArcFace [17] network, and $< \cdot, \cdot >$ represents the cosine similarity. Combining the above loss terms leads to our final objective, i.e.

$$\mathcal{L}_{\text{rec}} = \lambda_{\text{mse}}\mathcal{L}_{\text{mse}} + \lambda_{\text{perc}}\mathcal{L}_{\text{perc}} + \lambda_{\text{w}}\mathcal{L}_{\text{w}} + \lambda_{\text{id}}\mathcal{L}_{\text{id}}, \quad (8)$$

where λ_{mse}, λ_{perc}, λ_{w} and λ_{id} are the coefficients balancing different terms.

4. EXPERIMENTS

4.1. Degradation Model

To simulate real degenerated scenes, we use the following degradation model:

$$I_L = ((I_H * \mathbf{k}) \downarrow_s + \mathbf{n}_\sigma)_{JPEG_q}, \quad (9)$$

where $*$ represents the convolution operation between the HQ image I_H and the blur kernel $\mathbf{k}$. Note that the $\mathbf{k}$ is randomly selected from Gaussian blur, Motion blur, Average blur, and Motion blur with varying kernel sizes. In addition, $\downarrow_s$ is the downsampling operator with the scale factor s randomly selected in the range of $\left[\frac{64}{256}, 1\right]$. The $\mathbf{n}_\sigma$ denotes the Gaussian noise with a noise level σ, where σ is sampled from a normal distribution with $0 \leq \sigma \leq 25$, while $(\cdot)_{JPEG_q}$ is the JPEG compression operation with the quality factor q that is randomly chosen from $[5, 50]$.

4.2. Datasets

We built the HQ image set by randomly sampling W to generate 30,000 HQ images via $I_H = \mathcal{G}(W)$, rather than using the FFHQ dataset to train the model as in previous methods, encouraging the $\mathcal{E}(I_L)$ on the manifold of latent space. In the training phase, we resize all the images to 256×256 with bilinear downsampling and synthesize the LQ inputs with Eq. 9. Please notice that although the input size of our model is 256×256, our reconstruction result reaches 1024×1024. We use the CelebA-HQ dataset containing 30,000 images as the benchmark to compare our method with other state-of-the-art methods in both qualitative and quantitative experiments, on which we apply the degradation model expressed in Eq. 9 to synthesize LQ images imitating natural scene degeneration for evaluation.

4.3. Training Details

All of our models are implemented in PyTorch and optimized with Ranger optimizer, a combination of Rectified Adam [18] with the Lookahead technique [19], the parameters of which are set as $\beta_1 = 0.5$, $\beta_2 = 0.999$, and learning rate $lr = 0.0001$. We use the alternating optimize strategy to adapt the encoder and fusion module in the generator during the training phase, updating the fusion module every three encoder iterations. The trade-off parameters for different losses are empirically set as $\lambda_{\text{mse}} = \lambda_{\text{perc}} = 3$ and $\lambda_{\text{w}} = \lambda_{\text{id}} = 1$. We train our models on a single NVIDIA 2080ti GPU with a batch size of 4, consuming less than one day.

4.4. Comparisons

We compare our approach with other BFR methods, including HiFaceGAN [2], DFDNet [1], PSFR-GAN [3], PULSE [4], and GPEN [5], to evaluate performance. Because the input sizes of these methods are different, for comparison fairness, we resize the degraded images to the same size as used in the published models for inference.

As can be seen from Fig. 3, DFDNet, HiFaceGAN, and PSFR-GAN fail to produce reasonable face reconstructions, which tend to generate over-smoothed facial images. Furthermore, it is difficult for these approaches to deal with such complex degraded inputs, and the results are riddled with visible artifacts. PULSE achieves more realistic results than previously mentioned methods. However, it cannot faithfully recover the facial structure and loses most details in the background. Our method and GPEN are able to successfully restore more realistic textures in significantly degraded situations. For instance, in the third row and sixth column, the hair part of the recovery result by GPEN does not preserve the details because it skip-connects the features of all scales with the generator, resulting in existing artifacts in the LQ image being introduced into the result. It is worth noting that the optimization cost of our method is one order of magnitude cheaper than GPEN. The visual findings show that our technique can achieve more favorable visual outcomes, compared with the existing state-of-the-art methods, which explicitly reveals the advantages of our framework and training strategy for the BFR task.

As shown in Table 1, to further quantitatively evaluate our model, we compare with state-of-the-art approaches using the Peak Signal-to-Noise Ratio (PSNR), the Structural Similarity (SSIM), the Fréchet Inception Distances (FID), and the Learned Perceptual Image Patch Similarity (LPIPS). The best PSNR and SSIM metrics are obtained by directly upscaling the LQ image with the bilinear interpolation. However, such a simple operation cannot restore facial textures, indicating that PSNR and SSIM are inappropriate measures for the BFR task, same as the experiment in [5]. Alternatively, the FID and LPIPS indexes are better for reflecting the facial image perceptual quality. Since PULSE [4] is an optimization-based method, requiring lots of time to optimize for each LQ input, we only used the first 1,000 images of the CelebA-HQ dataset to calculate measures. In both FID and LPIPS, the proposed strategy outperforms DFDNet, HiFaceGAN, PSFR-GAN, and PULSE by a wide margin, demonstrating that our method can recover HQ images with excellent quality and fidelity. In comparison with GPEN, we reached the same result in the LPIPS index, while FID only slightly increases.

We further provide the comparison in terms of training



Fig. 3: Visual comparisons with other methods on CelebA-HQ dataset. **Zoom in for details please.**

Table 1: Quantitative comparisons with state-of-the-art BFR methods. (↑ means the higher score is preferred, and vice versa.) The best results are indicated in red, the second-best results are in blue.

Method	PSNR↑	SSIM↑	FID↓	LPIPS↓
Input	23.54	0.748	127.27	0.452
DFDNet [1]	22.61	0.519	36.74	0.372
HiFaceGAN [2]	17.41	0.667	47.69	0.375
PSFR-GAN [3]	17.78	0.642	22.91	0.345
PULSE [4]	20.95	0.641	48.50	0.405
GPEN [5]	21.86	0.668	12.99	0.314
Ours	21.14	0.675	13.28	0.314

cost between our method and GPEN [5] in Table 2. Because GPEN does not disclose the whole training duration in their framework, which includes two stages for generator learning and fine-tuning. As noticed by the authors, the fine-tuning phase of GPEN takes more than 14 days on a single P100 GPU. Retraining the generator with similar settings (config-f) of StyleGAN [11] consumed more than 69 days on a single V100 as reported officially, which is incredibly time-consuming. Our strategy only spends around 20 hours to converge. Furthermore, we attempted to increase the optimization steps, but trivial improvement on recovery performance was gained, indicating that our framework applies a simple and effective strategy to achieve the same or even better results as the previously designed methods.

Table 2: Comparison results in terms of training time on a single GPU card.

Method	Training Time	Device
StyleGAN [11]	>69 days	V100
GPEN [5]	>14 days	P100
Ours	**20 hours**	2080Ti

4.5. Ablation Study

In this section, we further verify the effectiveness of our proposed method on different configurations in Table 3. Figure 4 shows the visual results on different variants, which can also prove that higher PSNR and SSIM do not always mean better visual effects. Primarily, we investigate the effectiveness of the suggested fusion module as well as the latent space constraint. When the fusion module is discarded, the performance of all measures drops significantly. Furthermore, the visual result demonstrates that the encoder is hard to preserve the content of the degraded input without guidance. In other words, the feature-guided module can efficiently assist the generator for restoration. Despite both the PSNR and LPIPS decreasing when the latent space restriction is removed, the FID increases from 13.28 to 14.90, indicating that the restored images goes away from the distribution of the nature picture. We examine two modifications of our framework: fusing on larger feature maps (64×64) and employing multi-scale fea-

Fig. 4: Visual comparison between our variants of BFR. **RES**(n) denotes the resolution of feature maps is $n \times n$.

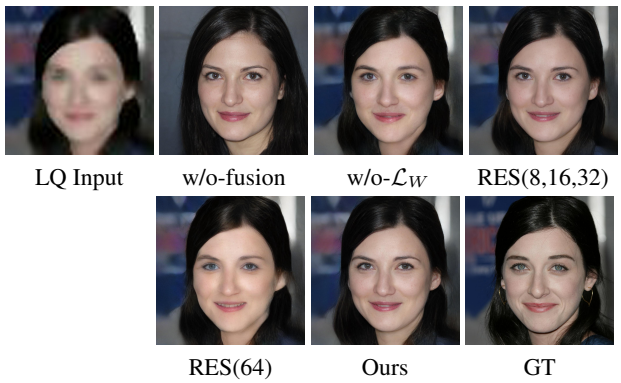


Table 3: Ablation study on different configurations.

Method	PSNR↑	SSIM↑	FID↓	LPIPS↓
w/o-fusion	15.92	0.600	25.68	0.420
w/o-$\mathcal{L}_w$	22.53	0.694	14.90	0.300
RES(8,16,32)	20.99	0.671	13.50	0.316
RES(64)	22.78	0.700	18.54	0.317
Ours	21.14	0.675	13.28	0.314

ture maps (from 8×8 to 32×32), to further evaluate the efficacy of our selected fusion strategy. The growth in PSNR and SSIM happens when larger feature maps are fed into the generator, however the results contain apparent artifacts. Employing multi-scale feature fusion degrades the performance since adding additional features to the generator will destroy feature distribution of the generator and hinder its generalization. Considering the number of parameters and training cost, fusion on a 32×32 size feature map seems to be a good choice. The above experimental results validate the effectiveness of our model and updating strategy.

5. CONCLUSION

This paper has explored the pre-trained GAN model to directly deal with the blind face restoration problem. We have proposed a network that leverages the pre-trained GAN with a feature fusion module to guide the restoration for coping with degradation facial images in complex scenarios. This research sheds new light on the potential to apply a learned generator to produce vibrant textures for degraded facial images, which can also be used in other domains. With significant save of training cost, our model can perform comparably to state-of-the-art methods both qualitatively and quantitatively. Experimental results reveal that one can directly utilize the existing GAN prior to synthesize high-quality textures for degraded images, which embraces broadly applicable scenarios.

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